Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/22/2015
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Synaptic plasticity in mammalian vestibular maculae		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	lfh@ucla.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9823
Organization Name:	University of California Los Angeles		
PI Address 1:	Geffen School of Medicine at UCLA		
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PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: End date is now 7/31/2014, per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:	July 2013 report: Dr. Ivan A. Lopez is added as a co-investigator.		
COI Name (Institution):	Zampighi, Guido (UCLA School of Medicine) Schweizer, Felix (University of California, Los Angeles) Lopez, Ivan (David Geffen School of Medicine at UCLA)		
Grant/Contract No.:	NNX09AL57G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The nervous systems of animals, including man, have been found to exhibit a remarkable ability to adapt to changes in the ambient gravity environment. An understanding of these adaptations is critical to managing human exploration and survivability during long-term spaceflight. Previous studies have shown that a gravity-sensing structure within the inner ear, the utricle, exhibited quantitative changes in components of its constituent hair cells during microgravity exposures up to 14 days. This structure is known as one of the vestibular maculae, inner ear structures with important load sensing capabilities. The changes that were observed involved modification in the number of synaptic ribbons, which are organelles involved in the transfer of sensory signals from the hair cells to neurons that communicate with the central nervous system. Such changes represent a form of adaptation that is more generally characterized as synaptic plasticity. At present, however, it is not known whether these changes represented a transient effect of short-term spaceflight, or whether they are more permanent and reflect a chronic adaptive modification. Therefore, the present study is designed to test whether the approximately 30-day exposure to microgravity proposed for the BION-M1 satellite mission results in chronic modification in synapse number within hair cells of the utricle. This will be achieved through transmission electron microscopy and quantitative analyses to estimate the density of synaptic ribbons in different hair cell types and in different regions of utricles from flight and control animals. In addition, we will utilize new methods of conical tomography to visualize these ribbon synapses in three dimensions. This enables us to test whether synaptic plasticity is manifested in changes in subpopulations of ultrastructural elements of the synaptic ribbon. These results will be critical to a more comprehensive understanding of the long-term exposure to spaceflight in mammalian gravity-sensing systems.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This research represents a high impact demonstration of the capabilities for vestibular hair cells to undergo synaptic plasticity due to alterations in the ambient gravitational field. This demonstration has led to a new investigation of hypergravity in which the mechanisms of this plasticity will be further explored. Eventually, these seminal investigations will segue to future studies of the molecular mechanisms underlying hair cell synaptic plasticity, which may potentially lead to the development of strategies for synaptic rehabilitation in pathologic conditions found in patients with compromised vestibular sensation. Such applications may be of significant value for vestibular rehabilitation following Meniere's disease or ototoxic damage due to chemotherapy.		
Task Progress:	We made significant progress in the processing, electron microscopy, imaging, and analysis of synaptic reconstructions via conical tomography. In addition, we implemented novel methods of serial electron microscopy, producing the first full images of presynaptic complexes in utricular hair cells from spaceflight and control specimens. The investigator team continues to collect and analyze ultrastructural data for additional tomograms and serial reconstructions. Two publications are in preparation, the first one reporting the synapse distribution data and the second on the ultrastructural findings from spaceflight specimens.		
Bibliography Type:	Description: (Last Updated: 10/05/2022)		
Abstracts for Journals and Proceedings	Lopez IA, Sheets K, Schweizer FE, Hoffman LF. "Tomographic analyses of afferent synapses in mouse utricular hair cells." Presented at the 37th Annual Midwinter Meeting of the Association for Research in Otolaryngology, San Diego, CA, February 22-26, 2014. Association for Research in Otolaryngology Abstracts. 2014;37: Abstract #PS-218. p. 116-117. , Feb-2014		

Abstracts for Journals and Proceedings	Hoffman LF, Lopez IA, Sheets K, Schweizer FE. "New insights of vestibular hair cell synaptic ultrastructure using TEM tomography and serial section SEM." Presented at the Society for Neuroscience Meeting, Neuroscience 2014, Washington, DC, November 15-19, 2014. Neuroscience 2014, Washington, DC, November 15-19, 2014. Program#/Poster#: 335.02/FF15. Available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=b68f238b-12e1-4ebc-b637-6dde521e343f&cKey=2937f30d-d693-4ab6-9d23-043f1e32e317&mKey=54c85d94-6d69-4b09-afaa-502c0e680ca7 ; accessed 1/22/2015. , Nov-2014
Abstracts for Journals and Proceedings	Hoffman LF, Lopez IA, Sheets K, Schweizer FE. "Advanced ultrastructural analyses of presynaptic complexes in utricular hair cells from BION•M1 mice." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Sultemeier DR, Choy KR, Schweizer FE, Hoffman LF. "Spaceflight-induced synaptic modifications within hair cells of the mammalian utricle." Journal of Neurophysiology. 2017 Jun 1;117(6):2163-78. Epub 2017 Feb 22. https://doi.org/10.1152/jn.00240.2016 ; PMID: 28228581 ; PMCID: PMC5454470 , Jun-2017